

Cross-provider interoperability for lunar positioning, navigation, and timing: a consistency-tolerance theorem and the irreducible inter-ephemeris orientation floor

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Abstract

A multi-provider lunar positioning, navigation, and timing (PNT) architecture—the declared vision of LunaNet, ESA Moonlight/LCNS, and their national counterparts—requires a number that no interoperability standard yet supplies: the *consistency tolerance*, the maximum cross-provider frame-and-dynamics disagreement a user can absorb while staying within a positioning budget. The LunaNet Interoperability Specification standardizes formats, frames, and time tags, but its cross-provider consistency tolerance is deferred to a still-unreleased applicable document—the Lunar Reference System and LunaNet Reference Time System Standard (AD-5, LNIS-TBD-AD0005). We supply it in three parts. First, we prove a *consistency-tolerance theorem*: to first order the pairwise inter-provider coordinate difference decomposes into a constant seven-parameter Helmert datum and a dynamics residual (uniquely where the sampled geometry gives the datum full column rank). A datum *common* to all providers is invisible to any single-provider user and cancels in provider-versus-provider fusion; it becomes real position error only against an absolute external tie. The object that a fusing or hand-off user actually pays—the “interop tax”—is instead the *differential* datum that two providers do not share. That tax splits, as a measured affine decomposition of the recovered rotation $\boldsymbol{\theta}_{\text{moon}} = \boldsymbol{\theta}_{\text{tie}} + \boldsymbol{\theta}_{\text{exc}}$, into a *reducible* part a common frame convention removes and an *irreducible* part that no coordinate convention removes; and inverting a triangle bound yields the closed-form tolerance $\tau(B)$ per Helmert parameter for a user budget B . Second, we anchor the budget on the only real “multiple providers” that exist today—the three independent authoritative lunar ephemerides DE440 (JPL), INPOP21a (IMCCE), and EPM2021 (IAA RAS)—and show, on their real 2024–2025 geocentric-Moon disagreement, that the cross-provider inconsistency is an *orientation* effect: a constant frame rotation explains 71–94% of the disagreement by amplitude (91–99.7% by variance), splitting into a reducible frame-tie ($\approx 1\text{--}3$ nrad) and an irreducible Moon-specific excess rotation ($\approx 5\text{--}6$ nrad for the JPL-versus-others pairs). Reported in the lever-arm-independent rotations, that irreducible excess is a *geocentric-Moon-state* inconsistency: at the Earth–Moon lever arm it is a ≈ 2 m envelope relevant to ephemeris hand-off and cislunar users, while the same excess rotation seen by a lunar-*surface* user is only ≈ 8 mm. This decomposition is VALIDATED: reproduced against an independent SciPy singular value decomposition to a relative and absolute tolerance below 10^{-3} . Third, we fold the two into a multi-provider interop budget and a design law: because the irreducible rotation is about twice the reducible (irreducible fraction $\|\boldsymbol{\theta}_{\text{exc}}\|/(\|\boldsymbol{\theta}_{\text{tie}}\| + \|\boldsymbol{\theta}_{\text{exc}}\|) = 0.687$), a common format-and-frame tag—what today’s standards mandate—leaves the dominant floor (a ≈ 1.8 m geocentric-Moon-state envelope at the Earth–Moon lever arm); closing it requires mandating a *common (or consistency-bounded) dynamical*

reference, not merely a common format. The scope AD-5 itself declares—a lunar gravity field to finite degree and order plus a parametric precession/nutation model—is exactly the lunar-orbit dynamics we identify as the irreducible floor. The theorem, gauge lemma, and tolerance inversion are proven; the inter-ephemeris decomposition is Validated on real data; the system-level budget and $\tau(B)$ map are explicitly MODELLED (the “providers” are ephemerides, a representative analog—no two lunar-PNT providers fly), and every number is reproducible from a single command.

Keywords: lunar PNT; interoperability; reference frame; datum realization; planetary ephemerides; DE440; INPOP; EPM; error budget; consistency tolerance; LunaNet; Moonlight

1 Introduction

The next decade will place several independent positioning, navigation, and timing (PNT) services around the Moon. NASA’s LunaNet and Lunar Communications Relay and Navigation Systems, ESA’s Moonlight programme and its Lunar Communications and Navigation Service (LCNS), and Japan’s Lunar Navigation Satellite System are each designed to broadcast navigation signals to surface and orbital users [6, 10]. The stated intent is *interoperability*: a user should be able to consume signals from more than one provider, or hand a position off between providers, and obtain a single consistent fix. Realizing that intent requires two distinct kinds of agreement. Providers must agree on *how* they talk—the message formats, the frame and time tags, the coordinate conventions—and they must agree on *what* they broadcast, closely enough that a user fusing them does not accumulate error from their disagreement.

The interoperability standards now being written address the first kind of agreement in detail and the second kind not at all. The LunaNet Interoperability Specification (LNIS) and the supporting CCSDS/IOAG cislunar work [9, 11] define lunar reference systems, frame realizations, timescales, and message formats; they make providers mutually *intelligible*. What they do not yet publish is a cross-provider *error budget*: the maximum amount by which two compliant providers may disagree—in frame realization and in broadcast dynamics—before a multi-provider user is pushed outside its accuracy requirement. In the applicable-document structure of the specification this quantity is deferred to AD-5, the Lunar Reference System and LunaNet Reference Time System Standard (LNIS-TBD-AD0005), which is not yet released and does not yet specify a cross-provider consistency tolerance. Tellingly, the scope AD-5 does declare—a lunar gravity field to finite degree and order together with a parametric precession/nutation model—is precisely the lunar-orbit dynamics that this paper identifies as the irreducible cross-provider floor. Supplying the number is hard because *no two lunar-PNT providers have flown*; there is, as yet, no measured cross-provider disagreement to budget against.

Our route around that obstacle is to observe that the “multiple providers” problem already exists in a real, measurable form one level down. Three independent institutions maintain authoritative, continuously updated lunar and planetary ephemerides using independent data reductions and dynamical models: DE440 from JPL/NASA [12], INPOP21a from IMCCE/Observatoire de Paris [5], and EPM2021 from IAA RAS [13]. Each realizes a lunar reference frame and a lunar orbit; each disagrees with the others by a real, decomposable amount. That disagreement is the closest existing analogue of a cross-provider inconsistency floor, and it can be measured today with no flight data whatsoever. We use it to anchor the budget, while stating plainly (throughout) that ephemerides are a representative analog for providers, not the providers themselves.

Related work. The lunar-PNT performance literature is, almost entirely, *user-side*: given a realized frame and a set of beacons, how accurately can a receiver be located. Pöhlmann et al. [14] derive error models and Cramér–Rao lower bounds for hybrid satellite/cooperative lunar navigation; open simulators such as LuPNT [8] target the same receiver-positioning question. These works take a single, given frame and dynamics as ground truth and do not model the error a user incurs from provider-to-provider disagreement. On the infrastructure side, the interoperability standards [9–11] and the reference-frame work define and realize the frames and timescales: Sośnica et al. [15] define and realize the International Lunar Reference Frame (ILRF) and quantify its origin-realization limit—the input, in our companion paper [3], to the per-provider frame-error covariance—while Fienga et al. [6] survey lunar reference systems, frames, and timescales in the Moonlight context, noting that ephemeris and frame differences matter for consistency. Our contribution is distinct from both strands: neither a user-side CRLB nor a frame realization, but the *cross-provider* budget that sits between them—how much two providers may differ before the user-side bounds are violated. We emphasize that the notion of an *irreducible* inter-provider floor is *derived* here from the structure of the real ephemeris disagreement, not quoted from any source; where prior surveys remark qualitatively that ephemeris differences are hard to remove, we make the statement precise, prove which part is removable by a frame convention and which is not, and measure both.

Contributions. We make three contributions, each with an explicit statement of what is proven, what is independently validated on real data, and what is representative.

1. **A cross-provider consistency-tolerance theorem** (PROVEN; Section 3). We formalize the interop tax as a free-network gauge phenomenon: a datum *common* to all providers is unobservable to a single-provider user and cancels in provider-versus-provider fusion (Lemma 1), becoming real position error only against an absolute external tie; the quantity a fusing or hand-off user actually pays is the *differential* datum δ_{ab} that two providers do not share. That differential tax splits—as a measured affine decomposition of the recovered rotation—into a *reducible* part removable by a common frame convention and an *irreducible* part removable only by a common dynamical reference (Lemma 2); and a triangle bound on the Helmert point-Jacobian inverts into a closed-form consistency tolerance $\tau(B)$ per Helmert parameter for a user budget B (Theorem 1). The tolerance $\tau(B)$ is exactly the number a standard at the AD-5 level must contain.
2. **The real inter-ephemeris decomposition** (VALIDATED; Section 4). We difference the DE440, INPOP21a, and EPM2021 geocentric-Moon states over 2024–2025 and decompose each pairwise disagreement into a Helmert datum and a residual, using the planetary (SSB) positions to discriminate a body-common frame-tie from a Moon-specific excess. We find the disagreement is an *orientation* effect—a constant rotation explains 71–94% by amplitude (91–99.7% by variance)—splitting into a reducible frame-tie ($\approx 1\text{--}3$ nrad) and an irreducible Moon-specific excess ($\approx 5\text{--}6$ nrad) for the JPL-versus-others pairs. Reported in the lever-arm-independent rotations, that excess is a geocentric-Moon-state inconsistency worth ≈ 2 m at the Earth–Moon lever arm (ephemeris hand-off and cislunar users) but only ≈ 8 mm for a lunar-surface user. This decomposition is VALIDATED against an independent SciPy singular value decomposition to a tolerance below 10^{-3} .
3. **An interop budget and a frame-versus-ephemeris design law** (MODELLED; Section 5). Folding the theorem and the anchor into a multi-provider budget, we show that under the

frame convention today’s standards pursue (agree on frame tags and format only) the irreducible rotation is about twice the reducible—irreducible fraction $\|\boldsymbol{\theta}_{\text{exc}}\|/(\|\boldsymbol{\theta}_{\text{tie}}\| + \|\boldsymbol{\theta}_{\text{exc}}\|) = 0.687$ —so a common format-and-frame tag leaves the dominant floor: a ≈ 1.8 m geocentric-Moon-state envelope at the Earth–Moon lever arm, closed only by mandating a common (or consistency-bounded) dynamical reference. We report the resulting $\tau(B)$ tolerance map for a lunar-surface user.

Honesty scope. We separate three kinds of claim. The *theorem*—the gauge lemma, the reducible/irreducible split as a decomposition of the differential tax, the triangle bound, and the $\tau(B)$ inversion form—is PROVEN; the measured *affine* attribution of the recovered rotation into a frame-tie and a Moon-excess is not (it is the modelling interpretation flagged below). The *inter-ephemeris decomposition*—the rotation-only Helmert-difference fit of the real DE440/INPOP21a/EPM2021 disagreement—is VALIDATED against an independent numerical oracle. The *system-level budget*: the multi-provider user model, the total interop budget, the design-law magnitudes, and the $\tau(B)$ map, are MODELLED, because the “providers” are ephemerides, a representative analog—no two lunar-PNT providers fly—and are tagged [M] wherever a system-level number appears. One further caveat runs throughout and is stated at each occurrence (Sections 3, 4, 7): attributing the planet-common rotation to a removable frame-tie is itself a modelling interpretation; the robust, convention-free claim is the Moon-*excess* rotation, which no whole-frame choice can remove. Section 6 makes the firewall precise, and every number is regenerated from the committed engine by a single command (Section 8).

2 The multi-provider datum-difference model

We model N providers, each of which realizes a lunar reference frame and a lunar ephemeris. For a single physical body-point $\mathbf{p}$ —a surface asset, or the lunocentre-referenced Moon state itself—provider a reports a coordinate $\mathbf{x}_a(\mathbf{p}, t)$. The object of study is the pairwise disagreement between two providers’ reports of the *same* physical point.

2.1 The seven-parameter datum difference

To first order, the difference between two providers’ coordinates for one point decomposes into a constant seven-parameter similarity (Helmert) datum and a time-varying residual,

$$\mathbf{x}_b(\mathbf{p}, t) - \mathbf{x}_a(\mathbf{p}, t) = \mathbf{J}(\mathbf{x}_a) \boldsymbol{\delta}_{ab} + \mathbf{e}_{ab}(\mathbf{p}, t), \quad (1)$$

where $\boldsymbol{\delta}_{ab} \in \mathbb{R}^7$ is the constant Helmert datum of b with respect to a , and $\mathbf{e}_{ab}$ is the residual that no single constant Helmert transform can absorb—the genuine differential *dynamics*, a prediction disagreement rather than a coordinate convention. The point-Jacobian is evaluated at the reporting provider’s coordinate $\mathbf{x}_a$; since $\mathbf{x}_a = \mathbf{p} + O(\|\boldsymbol{\delta}_{ab}\|)$, we may substitute the shared physical point $\mathbf{p}$, $\mathbf{J}(\mathbf{x}_a) = \mathbf{J}(\mathbf{p}) + O(\|\boldsymbol{\delta}_{ab}\|^2)$, so at the first order at which (1) holds the two are interchangeable. The decomposition is *unique* exactly when the sampled point geometry gives the 3×7 design a full column rank of seven; for a single lever arm it need not (a uniform radial scale and a radial translation are then indistinguishable), which is why the externally validated fit below drops the scale column (Section 2.2). The Helmert datum collects three translations $\mathbf{t} = (t_x, t_y, t_z)$, one scale s , and three small rotations $\boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$,

$$\boldsymbol{\delta} = (t_x, t_y, t_z, s, \theta_x, \theta_y, \theta_z)^\top \in \mathbb{R}^7, \quad (2)$$

and acts on a point $\mathbf{p} \in \mathbb{R}^3$ through the linearized similarity transform $\mathbf{p} \mapsto \mathbf{p} + \mathbf{t} + s\mathbf{p} + \boldsymbol{\theta} \times \mathbf{p}$, whose Jacobian with respect to $\boldsymbol{\delta}$ is the 3×7 point-Jacobian

$$\mathbf{J}(\mathbf{p}) = [\mathbf{I}_3 \mid \mathbf{p} \mid -[\mathbf{p}]_{\times}], \quad [\mathbf{p}]_{\times} \boldsymbol{\theta} = \mathbf{p} \times \boldsymbol{\theta}. \quad (3)$$

The first three columns are the translation, the fourth (the point’s own coordinates) is the scale, and the last three are the generators of rotation. This is the same point-Jacobian that governs the single-provider datum-identifiability problem of the companion paper [3]; here it acts on the *difference* between two providers rather than on a single frame’s null space.

Given a set of sampled points and epochs, the constant datum $\boldsymbol{\delta}_{ab}$ that best explains the differences (1) is the least-squares solution of the stacked linear system, and $\mathbf{e}_{ab}$ is the residual left after removing it. We solve this by centred normal equations with Jacobi (diagonal) preconditioning, for the reasons given in Appendix A.

2.2 The rotation-only sub-fit

The full seven-parameter fit is used only as an internal analytic check (Section 6). The externally validated decomposition uses a *rotation-only* sub-fit: the six-parameter rigid-body (Euclidean) model that retains translation and rotation but drops the scale column,

$$\mathbf{x}_b(\mathbf{p}, t) - \mathbf{x}_a(\mathbf{p}, t) = \mathbf{t}_{ab} - [\mathbf{p}]_{\times} \boldsymbol{\theta}_{ab} + \mathbf{e}_{ab}^{\text{rot}}(\mathbf{p}, t). \quad (4)$$

For a fixed body-point sampled through the physical geometry, the rotation $\boldsymbol{\theta}_{ab}$ is recovered from the centred point correspondences by the classical singular value decomposition of their cross-covariance [2]; the residual root-mean-square $\|\mathbf{e}_{ab}^{\text{rot}}\|$ measures the part of the disagreement that no constant translation-plus-rotation removes. We report, for each provider pair, the raw disagreement, the rotation-fit residual, and the recovered rotation magnitude; these are the VALIDATED quantities of Section 4. Using the rigid-body sub-fit rather than the full seven-parameter fit is deliberate: it is the minimal model whose recovered rotation is unambiguous (a scale column can trade against a radial translation, whereas a pure rotation cannot), so it is the fit we can reproduce against an independent oracle to a tight tolerance.

2.3 Reducible, irreducible, and the interop tax

Definition 1 (Interop tax). For a user that consumes coordinates from more than one provider (fusion) or ties a provider’s coordinate to an asset defined in another provider’s frame (hand-off, or a surveyed landmark), the *interop tax* is the position error induced by $\{\boldsymbol{\delta}_{ab}, \mathbf{e}_{ab}\}$ that a single self-consistent provider never sees.

Definition 2 (Reducible and irreducible components). Collect the pairwise disagreements (1). The recovered Moon rotation admits the measured affine split $\boldsymbol{\theta}_{\text{moon}} = \boldsymbol{\theta}_{\text{tie}} + \boldsymbol{\theta}_{\text{exc}}$, and the cross-provider error separates into three terms:

- a *reducible* part $\mathcal{R}$: the Helmert rotation $\boldsymbol{\theta}_{\text{tie}}$ that is common across bodies and directions and is therefore removable by adopting a common frame realization—a coordinate convention. Operationally we estimate it as the body-common (planet-common) frame-tie rotation shared by the Moon and the planetary positions;
- an *irreducible* part $\mathcal{I}$: the body-specific Moon-*excess* rotation $\boldsymbol{\theta}_{\text{exc}} = \boldsymbol{\theta}_{\text{moon}} - \boldsymbol{\theta}_{\text{tie}}$, a lunar-orbit orientation difference that does *not* appear in the planetary frame-tie. No coordinate convention removes $\mathcal{I}$; only a common designated dynamical reference (a common ephemeris) sets $\mathcal{I} \rightarrow 0$ by construction;

- a non-Helmert *residual* e_{ab} : the part of the difference that no constant rigid transform absorbs. It is bookkept separately from θ_{exc} (it is small, sub-nanoradian in rotation-equivalent), is likewise irreducible by any coordinate convention, and enters the budget total as its own term (Table 2).

The split $\mathcal{R}/\mathcal{I}$ is affine (not orthogonal): θ_{tie} and θ_{exc} are in general non-parallel, so their metre envelopes do not add in quadrature to the raw disagreement (Section 5).

The reducible/irreducible split is the load-bearing structure of the paper. Section 3 proves that a common frame convention removes only the body-common $\mathcal{R}$ and leaves $\mathcal{I}$ (and the residual) intact; Section 4 measures both parts on the real ephemerides; and Section 5 inverts the budget to the tolerance $\tau(B)$. We flag now, and repeat where it matters, that the *attribution* of the planet-common rotation to a reducible frame-tie (as opposed to shared dynamics) is a stated modelling interpretation; the convention-free claim is the Moon-excess rotation (Definition 2, second bullet), which is unambiguously Moon-specific and hence not removable by any whole-frame choice.

3 The consistency-tolerance theorem

This section states and proves the first contribution: a gauge lemma, a measured affine reducible/irreducible split with a proven removability structure, a closed-form consistency tolerance, and the design-law corollary. Each result carries an explicit honesty label. The algebraic statements (Lemma 1, the removability structure of Lemma 2, the triangle bound, and the $\tau(B)$ inversion form) are PROVEN; the affine *attribution* of the recovered rotation into a frame-tie and a Moon-excess is a modelling interpretation, stated as such (Remark 1); and the B_{eff} inflation and every system-level magnitude are MODELLED.

3.1 Gauge invisibility of a common datum

Lemma 1 (Gauge invisibility and the differential tax). *A datum component common to all providers— $\delta_a = \delta$ for every a —is unobservable to any single-provider user and, being common, also cancels in provider-versus-provider fusion, $\delta_{ab} = \delta_b - \delta_a = \mathbf{0}$. It becomes a real position error only against an absolute external tie: a landmark surveyed in a designated frame that does not carry the same gauge. The object a purely multi-provider (fusion or hand-off) user pays is instead the differential datum $\delta_{ab} \neq \mathbf{0}$ that two providers do not share—the interop tax, and the quantity Section 4 measures.*

Proof. A single provider’s internal range and PVT observables depend only on relative geometry: distances, delays, and clock differences between points expressed in that provider’s own frame. These are invariant under the frame *gauge*—the subspace of similarity transforms that the provider’s own observations cannot fix. For the single-frame lunar datum this residual gauge is generically of dimension at most six and collapses toward zero once lunar libration is observed and the scale becomes observable [3]; it is *not* the full seven-parameter similarity group. A common datum δ lying in that gauge changes no internal observable, so it is invisible intra-provider. Now form the cross-provider difference (1): a datum shared by both providers enters as $\delta_{ab} = \delta_b - \delta_a = \mathbf{0}$ and cancels identically—fusing or handing off between two providers that share a gauge costs nothing. The common gauge becomes a real position error only when a provider coordinate is tied to an *absolute* external reference that does not carry the same gauge, at which point it appears as $\mathbf{J}(\mathbf{p})\delta$; adopting a designated realization (one provider’s frame, or a designated ILRF realization [1, 15]) as the shared datum removes it by construction. The tax that a purely multi-provider user pays

is therefore not the common gauge but the *differential* datum $\delta_{ab} \neq \mathbf{0}$ of (1), the part the two providers do not share. $\square$

Lemma 1 separates two objects that are easily conflated: a *common* gauge, which cancels in multi-provider fusion and bites only against an absolute tie, and the *differential* datum δ_{ab} , which is invisible to each provider in isolation but real the instant two providers are mixed. The interop tax is the second. The lemma also identifies the mechanism by which a convention helps—agreement that shrinks δ_{ab} toward a shared realization—and thereby sets up the split.

3.2 The reducible/irreducible split

Lemma 2 (Reducible/irreducible split). *The cross-provider interop tax admits the measured affine split $\mathcal{R} + \mathcal{I} + \mathbf{e}$, where $\mathcal{R}$ (the reducible part) is the body-common Helmert rotation $\boldsymbol{\theta}_{\text{tie}}$, $\mathcal{I}$ (the irreducible part) is the body-specific Moon-excess rotation $\boldsymbol{\theta}_{\text{exc}} = \boldsymbol{\theta}_{\text{moon}} - \boldsymbol{\theta}_{\text{tie}}$, and $\mathbf{e}$ is the non-Helmert residual. Adopting a common frame realization sets $\mathcal{R} \rightarrow 0$ and leaves $\mathcal{I}$ and $\mathbf{e}$ unchanged; only adopting a common designated dynamical reference sets $\mathcal{I} \rightarrow 0$ (and drives $\mathbf{e} \rightarrow 0$ with it). The split is affine, not orthogonal: $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ are in general non-parallel.*

Proof. Write the pairwise tax from (1) as $\mathbf{J}(\mathbf{p})\delta_{ab} + \mathbf{e}_{ab}$. Split the recovered Moon rotation as $\boldsymbol{\theta}_{\text{moon}} = \boldsymbol{\theta}_{\text{tie}} + \boldsymbol{\theta}_{\text{exc}}$, where $\boldsymbol{\theta}_{\text{tie}}$ is the body-common frame-tie rotation estimated from the planetary positions (Section 4) and $\boldsymbol{\theta}_{\text{exc}} = \boldsymbol{\theta}_{\text{moon}} - \boldsymbol{\theta}_{\text{tie}}$ is the Moon-specific remainder; set $\mathcal{R} = \mathbf{J}(\mathbf{p})\boldsymbol{\theta}_{\text{tie}}$, $\mathcal{I} = \mathbf{J}(\mathbf{p})\boldsymbol{\theta}_{\text{exc}}$, and keep $\mathbf{e}_{ab}$ as a separate term. This is a measured *affine* decomposition, not an orthogonal one: $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ need not be perpendicular—on the real data they are 52° – 157° apart (Section 4)—so the metre envelopes $\|\mathcal{R}\|$ and $\|\mathcal{I}\|$ do not combine in quadrature to the raw disagreement. What the proof establishes is the *removability* structure. A common frame realization is a single similarity transform applied to every coordinate; it can cancel only a body-common rotation, so it sets $\mathcal{R} \rightarrow 0$ and leaves $\boldsymbol{\theta}_{\text{exc}}$ (hence $\mathcal{I}$) and the residual $\mathbf{e}_{ab}$ untouched— $\boldsymbol{\theta}_{\text{exc}}$ is by construction the part of the lunar rotation absent from the planetary tie, and $\mathbf{e}_{ab}$ is, in the aggregate least-squares sense $\sum_i \mathbf{J}(\mathbf{p}_i)^\top \mathbf{e}_{ab}(\mathbf{p}_i) = \mathbf{0}$, the component of the difference that no constant Helmert transform reaches. A common designated dynamical reference replaces each provider’s broadcast dynamics with one shared ephemeris, which by construction makes the differenced dynamics vanish and sets $\mathcal{I} \rightarrow 0$ (and $\mathbf{e}_{ab} \rightarrow 0$). $\square$

Remark 1 (The attribution is a modelling interpretation). Lemma 2 establishes the removability structure—which convention removes which part—and is PROVEN at that level. The affine *attribution* of the recovered rotation into a frame-tie and a Moon-excess, and which physical cause we assign to each, is a modelling choice. We identify the body-common rotation with a removable ICRF *frame-tie* (reducible) and the Moon-specific excess with irreducible *dynamics*. The convention-free, attribution-independent content is the Moon-*excess* rotation: it is by construction the part of the lunar disagreement that does *not* appear in the planetary frame-tie, so it is unambiguously Moon-specific and therefore removable by no whole-frame choice. The reader who declines the frame-tie interpretation may read the split as “planet-common rotation” versus “Moon-excess rotation” and lose only the word *reducible*, not the irreducible floor. We restate this caveat wherever the split is used (Sections 4, 5, 7).

3.3 The consistency tolerance

Let a mixed-provider user evaluate coordinates at lunocentric lever arm r (for a surface user, $r = R_{\text{Moon}}$; for the geocentric Moon state, r is the Earth–Moon distance). The worst-case interop-tax position error from a Helmert disagreement $\boldsymbol{\delta} = (\mathbf{t}, s, \boldsymbol{\theta})$ obeys a triangle bound.

Theorem 1 (Consistency tolerance). *For a body-point at radius $\|\mathbf{p}\| \leq r$,*

$$|\Delta \mathbf{x}(\mathbf{p})| = |\mathbf{J}(\mathbf{p})\boldsymbol{\delta}| \leq |\mathbf{t}| + |\mathbf{s}|r + |\boldsymbol{\theta}|r. \quad (5)$$

Consequently, to hold the interop tax within a budget B , the cross-provider consistency tolerances on the individual Helmert parameters are

$$|\mathbf{t}| \leq \tau_t = B_{\text{eff}}, \quad |\mathbf{s}| \leq \tau_s = \frac{B_{\text{eff}}}{r}, \quad |\boldsymbol{\theta}| \leq \tau_\theta = \frac{B_{\text{eff}}}{r}, \quad (6)$$

with $B_{\text{eff}} = B$ if the per-provider frame is realized exactly and $B_{\text{eff}} = \sqrt{\max(0, B^2 - \sigma_{\text{real}}^2)}$ otherwise, where σ_{real} is the single-provider datum-realization standard deviation from the companion frame-error covariance [3]. At the lunar lever arm the scale and rotation tolerances coincide at B_{eff}/r and are the binding (tightest) constraints; since r is large, an orientation error is the cheapest way to exhaust the budget.

Proof. From (3), $\mathbf{J}(\mathbf{p})\boldsymbol{\delta} = \mathbf{t} + \mathbf{s}\mathbf{p} + \boldsymbol{\theta} \times \mathbf{p}$. The triangle inequality gives $|\mathbf{J}(\mathbf{p})\boldsymbol{\delta}| \leq |\mathbf{t}| + |\mathbf{s}|\|\mathbf{p}\| + |\boldsymbol{\theta} \times \mathbf{p}|$, and $|\boldsymbol{\theta} \times \mathbf{p}| \leq |\boldsymbol{\theta}|\|\mathbf{p}\| \leq |\boldsymbol{\theta}|r$ while $|\mathbf{s}|\|\mathbf{p}\| \leq |\mathbf{s}|r$; substituting $\|\mathbf{p}\| \leq r$ yields (5). For the tolerance: if a single Helmert parameter is nonzero, the bound (5) contributes exactly that term, and requiring it not to exceed B_{eff} gives (6); the translation term is dimensionless in r while the scale and rotation terms carry the factor r , so at $r \gg 1$ the latter two bind first, at the common value B_{eff}/r . The inflation $B_{\text{eff}} = \sqrt{\max(0, B^2 - \sigma_{\text{real}}^2)}$ follows from budgeting in quadrature: a frame the provider cannot even realize consumes σ_{real} of the budget, leaving $\sqrt{B^2 - \sigma_{\text{real}}^2}$ for cross-provider agreement. $\square$

The triangle bound (5) and the inversion form (6) are PROVEN. Each tolerance in (6) bounds a *single* Helmert parameter *in isolation*: it is the largest that parameter may be if it alone is nonzero. If translation, scale, and rotation saturate their tolerances simultaneously, the triangle bound (5) permits a combined error of up to $3B_{\text{eff}}$. A joint budget guarantee therefore requires allocating B across the three terms— $B/3$ apiece for a hard split, or a quadrature allocation of the expected magnitudes—rather than granting each parameter the full B_{eff} ; (6) states the per-parameter ceilings, not a simultaneous one. The B_{eff} inflation is MODELLED: it depends on the companion CRLB σ_{real} , itself a modelled magnitude, and on treating realization and cross-provider errors as independent. When the per-provider realization is taken as exact (as in the reported tolerance map, Section 5), $B_{\text{eff}} = B$ and the tolerance is exact. The quantity $\tau(B)$ is precisely what an interoperability specification must state to convert a user PVT budget into a required per-provider frame-and-dynamics agreement: it is the number a standard at the AD-5 level—still unreleased—must contain.

3.4 The design-law corollary

Corollary 1 (Frame-versus-ephemeris design law). *Define the irreducible fraction of the cross-provider rotation budget as the linear ratio*

$$f_{\mathcal{I}} := \frac{\|\boldsymbol{\theta}_{\text{exc}}\|_{\text{rms}}}{\|\boldsymbol{\theta}_{\text{tie}}\|_{\text{rms}} + \|\boldsymbol{\theta}_{\text{exc}}\|_{\text{rms}}}. \quad (7)$$

Under the frame convention today’s interoperability standards pursue—agree on frame tags, format, and an ICRF realization, i.e. set $\mathcal{R} \rightarrow 0$ but not $\mathcal{I} \rightarrow 0$ —the residual interop tax is carried by $\boldsymbol{\theta}_{\text{exc}}$ (plus the small non-Helmert residual). If $f_{\mathcal{I}} > \frac{1}{2}$, the irreducible rotation exceeds the reducible one,

so a common frame/format convention leaves the dominant share of the cross-provider orientation disagreement in place; closing it requires additionally mandating a common (or consistency-bounded) designated dynamical reference, which sets $\mathcal{I} \rightarrow 0$.

Proof. Immediate from Lemma 2: adopting a common frame realization removes $\mathcal{R}$ and leaves $\boldsymbol{\theta}_{\text{exc}}$, so the post-convention orientation tax is $\|\boldsymbol{\theta}_{\text{exc}}\|$. The linear ratio (7) exceeds $\frac{1}{2}$ exactly when $\|\boldsymbol{\theta}_{\text{exc}}\|_{\text{rms}} > \|\boldsymbol{\theta}_{\text{tie}}\|_{\text{rms}}$. Because the split is affine rather than orthogonal (Lemma 2), we compare the rotations directly and do not combine them in quadrature; the lever-arm-independent rotations are the primary quantities, with metres following as worst-case envelopes at a stated lever arm (Section 5). Removing $\mathcal{I}$ requires the differenced dynamics to vanish, which a common designated ephemeris achieves by construction. $\square$

Corollary 1 is a proven implication of the split; its *premise*—whether the irreducible fraction actually exceeds $\frac{1}{2}$ for real lunar providers—is an empirical question answered by the inter-ephemeris anchor of the next section, where we measure an irreducible fraction of 0.687 (the irreducible rotation is about twice the reducible) for the representative provider set. That the premise holds, and hence that a common format leaves a real floor, is the design law’s content; the specific magnitude of the floor is MODELLED (Section 5).

4 The real inter-ephemeris disagreement

We now anchor the budget on the real disagreement between the three independent authoritative lunar ephemerides. This is the paper’s one VALIDATED result: the rotation-only decomposition of the measured DE440/INPOP21a/EPH2021 disagreement, reproduced against an independent numerical oracle.

4.1 Method

We sample the geocentric Moon position (NAIF body 301 with respect to 399) from each of DE440 [12], INPOP21a [5], and EPH2021 [13] over 2024-01-01 to 2025-12-31 in Barycentric Dynamical Time (TDB), at two-day cadence (366 epochs), in the ICRF/J2000 frame, using the IMCCE `calceph` library so that all three families are read through one interface (Appendix A). For each provider pair we form the epoch-wise position difference and fit the rotation-only (six-parameter) rigid-body model (4) by the singular value decomposition of the centred cross-covariance [2], recording the raw disagreement RMS, the rotation-fit residual RMS, and the recovered Moon rotation magnitude $|\boldsymbol{\theta}_{\text{moon}}|$.

To separate a body-common frame-tie from a Moon-specific excess we additionally sample the planetary positions with respect to the Solar System barycentre (Mercury, Venus, Mars, and the Earth–Moon barycentre; NAIF bodies 1, 2, 4, 3 with respect to 0). We fit the same rotation-only model separately to each of these four witness bodies, obtaining four per-body rotation vectors, and take $\boldsymbol{\theta}_{\text{tie}}$ as their *component-wise median*: a deliberately robust but ad-hoc aggregate that resists a single discrepant witness (a joint or weighted fit would shift the split; Section 7.3 and Appendix A). The Sun is deliberately *excluded*: its Solar-System-barycentre position is dominated by the giant planets and is barycentre-definition-contaminated, so it is not a clean witness of the ICRF frame-tie (Appendix A). The Moon-specific *excess* rotation is then $\boldsymbol{\theta}_{\text{exc}} = \boldsymbol{\theta}_{\text{moon}} - \boldsymbol{\theta}_{\text{tie}}$. We report the rotations—which are lever-arm-independent—as the primary quantities; the metre columns are worst-case envelopes obtained by mapping each rotation through the Earth–Moon lever arm $r = 3.840 \times 10^8$ m, so the reducible (frame-tie) envelope is $|\boldsymbol{\theta}_{\text{tie}}| r$ and the irreducible (Moon-orbit

Table 1: The real inter-ephemeris disagreement (2024–2025 geocentric Moon, two-day cadence, 366 epochs), decomposed by the rotation-only Helmert-difference fit. The rotations (right three columns, nrad) are the lever-arm-independent primary quantities; the reducible and irreducible metre columns are worst-case envelopes $|\boldsymbol{\theta}_{\text{tie}}| r$ and $|\boldsymbol{\theta}_{\text{exc}}| r$ at the geocentric-Moon-state lever arm $r = 3.840 \times 10^8 \text{ m}$ (upper bounds; because $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ are non-parallel they do not sum to the raw disagreement). This decomposition is VALIDATED against an independent SciPy singular value decomposition to a relative and absolute tolerance below 10^{-3} (Section 6). The full seven-parameter Helmert fit is a separate internal analytic check and is *not* claimed to be externally validated.

Pair	raw (m)	rot. res. (m)	reducible (m)	irreducible (m)	$ \boldsymbol{\theta}_{\text{moon}} $ (nrad)	$ \boldsymbol{\theta}_{\text{tie}} $ (nrad)	$ \boldsymbol{\theta}_{\text{exc}} $ (nrad)
DE440–INPOP21a	2.3955	0.1387	0.8845	1.8741	6.5598	2.3034	4.8804
DE440–EPM2021	2.0148	0.2789	0.4288	2.4056	5.2567	1.1166	6.2646
INPOP21a–EPM2021	0.7236	0.2124	1.0209	0.5884	2.4171	2.6585	1.5323

dynamics) envelope is $|\boldsymbol{\theta}_{\text{exc}}| r$. Because $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ are in general non-parallel, these envelopes are upper bounds that do not sum to the raw disagreement. This lever arm is the *geocentric-Moon state*: the metres quantify a Moon-state (ephemeris hand-off / cislunar) inconsistency, not a surface-user error; a lunar-surface user at radius R_{Moon} sees the same rotations scaled by the far smaller R_{Moon} (Section 4.2).

4.2 Result

Table 1 reports the decomposition for the three provider pairs. Three facts stand out, and all three are properties of the real data.

The disagreement is an orientation effect. A constant frame rotation removes the overwhelming majority of the disagreement: the rotation-fit residual falls to 0.14–0.28 m from a raw disagreement of 0.72–2.40 m. Quantifying the rotation’s share honestly requires care about the metric. By *amplitude*, $(\text{raw} - \text{rot. res.})/\text{raw}$, rotation explains 94% (DE440–INPOP21a), 86% (DE440–EPM2021), and 71% (INPOP21a–EPM2021), i.e. 71–94%; by *variance*, $1 - (\text{rot. res.}/\text{raw})^2$, it explains 99.7%, 98%, and 91%, i.e. 91–99.7%. We report both metrics and their ranges rather than a single headline figure. The disagreement is emphatically *not* a translation, a scale, or a format difference—those remove almost nothing—and the practical consequence is that a common frame/format tag, which can only cancel a body-common part of the rotation, is attacking the right quantity but not all of it.

The rotation splits, and for the JPL pairs the irreducible part dominates. The recovered Moon rotation separates, via the measured affine split $\boldsymbol{\theta}_{\text{moon}} = \boldsymbol{\theta}_{\text{tie}} + \boldsymbol{\theta}_{\text{exc}}$, into a body-common frame-tie ($|\boldsymbol{\theta}_{\text{tie}}| = 1.1\text{--}2.7$ nrad; reducible) and a Moon-specific excess ($|\boldsymbol{\theta}_{\text{exc}}| = 1.5\text{--}6.3$ nrad; irreducible). The two are *non-parallel*—the angle between $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ is 52° , 157° , and 116° for the three pairs—so the split is affine, not orthogonal, and the metre envelopes below are upper bounds that do not add in quadrature. For the two JPL-versus-others pairs the irreducible excess rotation is 4.88 nrad (DE440–INPOP21a) and 6.26 nrad (DE440–EPM2021), each roughly twice the reducible frame-tie (2.30 and 1.12 nrad); at the geocentric-Moon-state lever arm these are irreducible envelopes of 1.87 m and 2.41 m against reducible envelopes of 0.88 m and 0.43 m. The

INPOP21a–EPM2021 pair, whose families share more heritage, is the exception and is more balanced (2.66 nrad tie versus 1.53 nrad excess; 1.02 m versus 0.59 m). As an internal cross-check, the DE440–INPOP21a excess-rotation vector is (0.601, 3.362, -3.486) nrad, of magnitude 4.880 nrad, which maps to the 1.874 m irreducible envelope at the lever arm. These metres are geocentric-Moon-state inconsistencies (relevant to ephemeris hand-off and cislunar users); the identical excess rotations experienced by a lunar-*surface* user scale instead by $R_{\text{Moon}} = 1.7374 \times 10^6$ m, so the RMS irreducible excess (4.67 nrad, Section 5) is only ≈ 8 mm at the surface and even the largest single-pair excess (6.26 nrad) is ≈ 11 mm.

The order of magnitude is physically reasonable, and is corroborated by the literature.

A metre-level geocentric-Moon disagreement between DE440 and INPOP21a is consistent with the known level of agreement between these independent reductions, which supports the correctness of the *calceph* reads underlying the fit. Independent corroboration comes from Sośnica et al. [15], who compare the same ephemeris families against their ILRF combined solution and report along-track RMS lunar-position differences of ≈ 0.51 m (EPM2021), ≈ 0.60 m (INPOP21a), and ≈ 1.67 m (DE430), against much smaller radial (0.08–0.17 m) and cross-track (0.21–0.37 m) differences, and attribute the dominant along-track disagreement to dynamics (tidal dissipation and external torques). This is order-of-magnitude support for *both* of our claims—the ≈ 2 m magnitude and the irreducible-*dynamics* attribution—from an entirely independent comparison. We stress it is not a validation target: their comparison is DE430 (we use DE440) and pairwise-versus-combined (we compare providers pairwise), so it corroborates the order of magnitude and the dynamical origin, not a specific number. We do not otherwise claim to reproduce any published inter-ephemeris comparison as a target; the decomposition, not the raw magnitude, is the validated object.

4.3 What is validated, and what is not

The rotation-only decomposition of Table 1—raw, rotation residual, reducible, irreducible, and the three rotation magnitudes, for all three pairs—is reproduced by the engine and by an independent SciPy singular value decomposition of the same six-decimal fixture data, agreeing to a relative error below 10^{-3} and an absolute error below 10^{-3} m (metre quantities) and 10^{-2} nrad (angle quantities), even for the near-cancelling INPOP21a–EPM2021 excess where $\theta_{\text{moon}} \approx \theta_{\text{tie}}$. This is the paper’s single VALIDATED row (Section 6). We stress two boundaries. First, only the *rotation-only* decomposition is externally validated; the full seven-parameter Helmert fit (which additionally estimates translation and scale) is exercised only by an internal analytic-recovery unit test and is not claimed to be externally validated. Second, the reducible/irreducible *attribution*—calling the planet-common rotation a removable frame-tie—is the modelling interpretation of Remark 1; the convention-free content is the Moon-excess rotation column $|\theta_{\text{exc}}|$, which is unambiguously Moon-specific and removable by no whole-frame choice.

5 The interop budget and the design law

We now fold the theorem (Section 3) and the real anchor (Section 4) into a multi-provider interoperability budget and its consistency-tolerance map. Every quantity in this section is MODELLED and tagged [M]: the “providers” are the three ephemerides, a representative analog for a real multi-provider constellation—*no two lunar-PNT providers fly*—so the budget totals and tolerances describe the structure and ordering of the trade, not a certified system budget.

Table 2: Multi-provider interop budget (MODELLED, [M]; the “providers” are the three ephemerides, a representative analog). The reducible and irreducible columns are RMS-combined worst-case lever-arm envelopes $|\boldsymbol{\theta}|r$ at $r = 3.840 \times 10^8$ m (upper bounds; non-additive, since $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ are non-parallel). The **PerProvider** total is the *measured raw RMS* disagreement (1.855 m), not a quadrature sum; **CommonFrameTie** removes the reducible frame-tie and leaves an irreducible-dominated ≈ 1.8 m geocentric-Moon-state floor; only **CommonEphemeris** zeroes it. The irreducible fraction $\|\boldsymbol{\theta}_{\text{exc}}\|/(\|\boldsymbol{\theta}_{\text{tie}}\| + \|\boldsymbol{\theta}_{\text{exc}}\|) = 0.687$ (a linear ratio of the RMS rotations) is convention-independent. The three pairwise differences are linearly dependent $((A-B) + (B-C) = (A-C))$, so the RMS combination overcounts—a MODELLED choice.

Convention	reducible env. (m)	irreducible env. (m)	total (m)	irreducible fraction
PerProvider	0.8182	1.7931	1.8550	0.6867
CommonFrameTie	0.8182	1.7931	1.8062	0.6867
CommonEphemeris	0.8182	1.7931	0.0000	0.6867

5.1 The budget under three frame conventions

Combining the three provider pairs of Table 1 gives a representative cross-provider position-inconsistency budget, evaluated under three interoperability conventions of increasing strength: (i) **PerProvider**, each user keeps each provider in its own realized frame and dynamics; (ii) **CommonFrameTie**, users adopt a common frame realization, removing the reducible frame-tie $\mathcal{R}$ but not the irreducible dynamics $\mathcal{I}$ —this is what today’s format-and-frame standards achieve; and (iii) **CommonEphemeris**, users adopt a common designated dynamical reference, removing $\mathcal{I}$ as well. Table 2 reports the result. Following the paper’s convention, the rotations are the primary quantities and metres are envelopes: the reducible and irreducible columns are the RMS-combined frame-tie (0.8182 m, from $|\boldsymbol{\theta}_{\text{tie}}|_{\text{rms}} = 2.131$ nrad) and Moon-excess (1.7931 m, from $|\boldsymbol{\theta}_{\text{exc}}|_{\text{rms}} = 4.669$ nrad) *worst-case envelopes* $|\boldsymbol{\theta}|r$ at the Earth–Moon lever arm. The **PerProvider** total is the *measured raw RMS* disagreement (1.855 m), not the sum of the two envelopes, because $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ are non-parallel and the envelopes do not add in quadrature; each convention then zeroes the components it can remove. One further caveat is MODELLED: the three pairwise differences are linearly dependent $((A-B) + (B-C) = (A-C))$, so RMS-combining them overcounts the independent information—a deliberate, stated modelling choice for a representative envelope.

The design law follows directly. The irreducible fraction is 0.687: the irreducible Moon-excess rotation is about twice the reducible frame-tie, so more than two-thirds of the cross-provider orientation budget lives in the part that no coordinate convention removes. Adopting a common frame tie—agreeing on frame tags, format, and an ICRF realization—lowers the geocentric-Moon-state budget only from the measured 1.855 m to 1.8062 m, leaving a ≈ 1.8 m irreducible-dominated floor (a Moon-state inconsistency at the Earth–Moon lever arm; the same excess rotation is only ≈ 8 mm for a lunar-surface user); only a common designated dynamical reference drives the total to zero. The budget is monotone in convention strength, **CommonEphemeris** $\leq$ **CommonFrameTie** $<$ **PerProvider**, as Corollary 1 requires. This is the concrete, real-data-backed input to an interoperability specification: *a common format and frame realization is necessary but not sufficient; the cross-provider floor is set by the lunar-orbit dynamics, and closing it requires mandating a common (or consistency-bounded) ephemeris.*

Table 3: Consistency tolerance $\tau(B)$ (MODELLED, [M]) for a lunar-surface user at $r = 1\,737\,400$ m, from the triangle bound of Theorem 1 with $B_{\text{eff}} = B$. The origin tolerance is B ; the scale and rotation tolerances coincide at B/r and are the binding (tightest) constraints—an orientation error is the cheapest way to exhaust the budget at the lunar lever arm.

B (m)	max origin (m)	max scale	max rotation (rad)	max rotation (nrad)
5.0	5.0000	2.8779×10^{-6}	2.877863×10^{-6}	2877.86
10.0	10.0000	5.7557×10^{-6}	5.755727×10^{-6}	5755.73
15.0	15.0000	8.6336×10^{-6}	8.633590×10^{-6}	8633.59

5.2 The consistency tolerance map

Inverting Theorem 1 for a lunar-surface user at $r = R_{\text{Moon}} = 1\,737\,400$ m, with the per-provider realization taken as exact ($B_{\text{eff}} = B$), gives the consistency tolerances of Table 3: the maximum inter-provider disagreement, per Helmert parameter, that keeps a mixed-provider surface user within a position budget B . Note the lever arm here is the lunar *surface* radius R_{Moon} , two orders of magnitude smaller than the Earth–Moon lever arm at which the ≈ 1.8 m geocentric–Moon-state floor of Table 2 is evaluated; the two tables therefore speak to different users (a surface asset versus an ephemeris/cislunar hand-off) and must not be compared metre-for-metre.

For every budget the binding constraint is the rotation (equivalently the scale): at the lunar radius a rotation of 2877.86 nrad already displaces a surface asset by the full 5 m, whereas the origin can drift the full 5 m before it binds. This is consistent with, and explains the practical importance of, the real anchor being orientation-dominated (Section 4): the disagreement that actually exists between lunar providers is exactly of the kind—orientation—to which a surface user’s budget is most sensitive. A specification writer can read Table 3 as the required cross-provider rotation agreement: to guarantee a 5 m mixed-provider surface budget, compliant providers must agree in orientation to better than $\approx 2.9 \mu\text{rad}$. Whether they do so today is the question the anchor answers negatively for the irreducible Moon-excess: the measured excess rotations (1.5–6.3 nrad, Table 1) are nearly three orders of magnitude *below* the 2877.86 nrad tolerance (a factor of 460–1900), so at a 5 m budget the current ephemeris-level orientation disagreement is not itself budget-threatening; the map’s value is that it sets the tolerance a real provider constellation must not exceed, and identifies orientation as the parameter to watch.

5.3 Application to LNIS AD-5 and multi-provider procurement

Two immediate uses follow, both stated as MODELLED guidance rather than a certified budget. First, Table 3 supplies the functional form and representative magnitude of the cross-provider consistency tolerance that AD-5—the still-unreleased Lunar Reference System and LunaNet Reference Time System Standard (LNIS-TBD-AD0005)—does not yet specify: a user PVT budget B maps to a per-parameter cross-provider tolerance, binding on orientation, $\tau_\theta = B/R_{\text{Moon}}$. The alignment is pointed: the scope AD-5 itself declares—a lunar gravity field to finite degree and order together with a parametric precession/nutation model—is exactly the lunar-orbit dynamics that our anchor identifies as the irreducible floor, so a standard at the AD-5 level is already scoped over the very term that dominates the budget. Second, Table 2 gives a vendor-neutral design law for a multi-provider interoperability framework such as LunaNet, or an ESA procurement such as Moonlight/LCNS: mandating a common message format and frame realization is necessary but removes only the sub-metre reducible part; a residual floor set by the providers’ independent lunar-orbit dynamics remains, and a programme that seeks metre-consistent multi-provider PNT must addi-

tionally require a common designated ephemeris, or bound the permissible ephemeris divergence, as an explicit interoperability criterion. We reiterate the honesty boundary: the ephemerides stand in for providers, the budget totals and tolerances are representative, and the reducible/irreducible attribution carries the caveat of Remark 1. The convention-free content is robust: an irreducible, orientation-type, Moon-specific disagreement exists between the real lunar dynamical references, and no coordinate convention removes it.

6 Validation and the structure/magnitude firewall

A paper that converts a metre-level lunar disagreement into a procurement design law must be scrupulous about which of its numbers are earned on real data and which are representative. We enforce this with an explicit firewall, implemented as a rule in the engine’s verification suite and stated plainly here: a quantity is **VALIDATED** only when it is checked against an *independent* oracle—a separate dataset, library, or published vector—to a stated tolerance; a result is **PROVEN** when it is an algebraic theorem; everything else is **MODELLED**. This section states what falls on each side and gives the external check that anchors the **VALIDATED** side.

6.1 The independent oracle for the decomposition

The one load-bearing empirical object of this paper is the rotation-only decomposition of the real inter-ephemeris disagreement (Table 1). We validate it against an independent SciPy/NumPy reference. A generator script (Section 8) reads the DE440/INPOP21a/EPM2021 kernels through **calceph**, samples the geocentric Moon and the planetary SSB positions on the same grid used by the engine, and computes the rotation-only Helmert-difference fit by an independent singular value decomposition of the centred cross-covariance [2], emitting the expected raw disagreement, rotation residual, reducible and irreducible metres, and the three rotation magnitudes to a vendored fixture (**reference.json**). Crucially, both the engine and the oracle consume positions rounded to six decimals, so the comparison is byte-consistent and tests the *algorithm*, not floating-point sampling noise (Appendix A). The engine’s **provenance_split** reproduces every value to a relative error below 10^{-3} and an absolute error below 10^{-3} m (metre quantities) and 10^{-2} nrad (angle quantities)—including the near-cancelling INPOP21a–EPM2021 excess, where θ_{moon} and θ_{tie} nearly coincide and the subtraction is the numerically hardest. This is the paper’s single **VALIDATED** row: the decomposition of the *real* ephemeris disagreement is externally confirmed by an independent implementation.

6.2 What is Validated, Proven, and Modelled

Table 4 draws the line explicitly. Only the rotation-only inter-ephemeris decomposition is **VALIDATED**. The gauge lemma, the removability structure of the reducible/irreducible split, the triangle bound, and the $\tau(B)$ inversion are **PROVEN** (algebra); the affine numeric attribution of the rotation into a frame-tie and a Moon-excess is a modelling interpretation (Remark 1). Everything at the system level—the multi-provider user model, the budget totals of Table 2, the design-law magnitudes, and the $\tau(B)$ map of Table 3—is **MODELLED**, because the “providers” are ephemerides, a representative analog. Two internal checks are honest but not independent and are labelled so: the full seven-parameter Helmert fit is exercised only by an analytic-recovery unit test (a synthetic datum is injected and recovered), and the budget assembly is an internal RMS combination of the validated pairwise numbers.

Table 4: The structure/magnitude firewall. Only the rotation-only inter-ephemeris decomposition is VALIDATED (independent SciPy/NumPy oracle on real DE440/INPOP21a/EPM2021 data). The theorem is PROVEN; every system-level magnitude is MODELLED. The full seven-parameter Helmert fit and the budget assembly are internal-consistency checks, honest but not independent.

Result	Check	Oracle	Status
Rotation-only inter-ephemeris decomposition (Table 1)	real DE440/INPOP/EPM vs SciPy SVD, $<10^{-3}$	ext. library + real data	VALIDATED
Gauge lemma; R/I removability structure; triangle bound; $\tau(B)$ inversion (Lem. 1–2, Thm. 1)	algebraic proof	—	PROVEN
Full 7-parameter Helmert fit	analytic datum injection/recovery	internal	MODELLED
Multi-provider interop budget (Table 2)	RMS assembly of validated pairs	internal	MODELLED
Consistency-tolerance map (Table 3)	triangle bound, $B_{\text{eff}} = B$	internal (analytic)	MODELLED
B_{eff} inflation (Thm. 1)	quadrature with companion CRLB	internal (mod. σ)	MODELLED

6.3 The split caveat, restated

The firewall has one boundary that is a modelling interpretation rather than a tolerance, and we restate it here so it is impossible to miss. Attributing the planet-common rotation to a removable ICRF frame-tie (the *reducible* part) and the Moon-excess to irreducible dynamics is a stated modelling interpretation (Remark 1). It is the reasonable reading—a rotation shared by the Moon and the planets is naturally a frame-tie—but it is not forced by the data. The attribution-independent, convention-free claim, the one a skeptical reviewer can accept without granting the interpretation, is the Moon-*excess* rotation $|\boldsymbol{\theta}_{\text{exc}}|$: it is by construction the part of the lunar disagreement that does not appear in the planetary tie, hence unambiguously Moon-specific and removable by no whole-frame choice. Every irreducible-floor and design-law statement in the paper rests only on the excess, not on the frame-tie label.

6.4 Reproducibility

Every quantitative result in this paper—the three-pair decomposition of Table 1, the budget of Table 2, and the tolerance map of Table 3—is regenerated from the committed engine and the vendored fixtures by a single command (Section 8). The VALIDATED row is asserted in the continuous-integration test suite, which enforces the external-oracle gate: the pairwise decomposition must match `reference.json` to the stated tolerance or the build fails. There are no hand-entered results in the tables: the manuscript numbers and the engine outputs are the same numbers, traceable to one run.

7 Discussion

7.1 Implications for interoperability standards

The practical import of the paper is one design law: *a common format and frame is necessary but not sufficient; a metre-consistent multi-provider lunar PNT architecture must additionally mandate a common (or consistency-bounded) dynamical reference*. Today’s interoperability work—LNIS, the CCSDS/IOAG cislunar standards, LunaNet [9–11]—standardizes the parts of the disagreement that are, in our decomposition, *reducible*: the message formats, the frame tags, and (through a designated ILRF realization [15]) the common frame-tie. That work is essential and it attacks the right quantity, orientation. But the real inter-ephemeris anchor shows the disagreement is dominated by an *irreducible* Moon-specific orientation difference that no frame convention removes (irreducible fraction 0.687: the irreducible rotation is about twice the reducible, Table 2); a format-and-frame standard alone therefore leaves a ≈ 1.8 m geocentric-Moon-state floor at the Earth–Moon lever arm (a few millimetres for a surface user). Our contribution to the still-unreleased AD-5—the Lunar Reference System and LunaNet Reference Time System Standard (LNIS-TBD-AD0005), which does not yet specify a cross-provider consistency tolerance—is the functional form and representative magnitude of that tolerance (Table 3), binding on orientation at $\tau_\theta = B/R_{\text{Moon}}$; and it is doubly apposite because AD-5’s own declared scope (a lunar gravity field to finite degree and order plus a parametric precession/nutation model) is exactly the lunar-orbit dynamics we identify as the floor. Our contribution to a vendor-neutral programme—a multi-provider framework such as LunaNet, or an ESA procurement such as Moonlight/LCNS—is the corresponding requirement structure: either a common designated lunar ephemeris, or an explicit bound on permissible ephemeris divergence, alongside the format and frame requirements.

7.2 Relation to prior work

This paper sits between two established strands and connects them. On one side, the user-side lunar-PNT performance analyses [8, 14] bound the accuracy of locating a receiver given a single realized frame and dynamics; they do not model provider-to-provider disagreement. Our budget supplies exactly the missing input to those analyses: the cross-provider error a fusion or hand-off user incurs, on top of the single-provider CRLB, when the frame and dynamics are not shared. On the other side, the frame-realization work [6, 15] defines and realizes the ILRF and its timescales; our companion paper [3] classifies the identifiability of that single-provider datum and supplies the per-provider frame-error covariance σ_{real} that enters the B_{eff} inflation of Theorem 1. The present work is the cross-provider layer above both: it takes each provider’s realized frame and dynamics as given and budgets their mutual disagreement. The gauge lemma (Lemma 1) makes the connection to the companion paper precise—the single-provider datum gauge (generically of dimension at most six, and collapsing under libration as the scale becomes observable) is the object whose *differential* across providers becomes the interop tax; a gauge *common* to both providers, by contrast, cancels in fusion and becomes real error only against an absolute external tie.

7.3 Limitations

The firewall of Section 6 is also a statement of limitations, and we make them explicit.

- **The providers are ephemerides.** The single most important caveat: no two lunar-PNT providers have flown, so the three ephemerides are a *representative analog* for a real multi-provider constellation. The budget totals (Table 2) and the tolerance map (Table 3) are MODELLED; only the pairwise decomposition (Table 1) is VALIDATED. A real provider constellation

would disagree for additional reasons (broadcast-ephemeris fit error, clock and signal-in-space errors, regional frame ties) not present in the ephemeris comparison.

- **The reducible/irreducible attribution is a modelling interpretation.** As stated in Remark 1 and Section 6.3, calling the planet-common rotation a removable frame-tie is a reasonable reading, not a forced one. All irreducible-floor and design-law claims rest only on the convention-free Moon-excess rotation.
- **The frame-tie estimator is a robust but ad-hoc aggregate.** The body-common rotation is estimated as the *component-wise median* of four separate per-witness-body rotation-only fits (Mercury, Venus, Mars, and the Earth–Moon barycentre), with the Sun excluded as barycentre-contaminated (Appendix A). The median is deliberately robust to a single discrepant witness, but it is not the only reasonable choice: a different witness set, or a different aggregation (a joint or a weighted fit), would shift the reducible/irreducible split quantitatively. The qualitative finding—a dominant Moon-specific excess for the JPL-versus-others pairs—is robust to reasonable choices, but the exact metre split is not a validated invariant.
- **The budget model is first-order and static.** The datum difference (1) is a first-order Helmert model with a constant datum over the two-year window; it does not resolve time-dependence of the datum, higher-order terms, or the along-track structure of the residual. These would refine the residual but do not change the orientation-dominated structure that the rotation fit already captures at the 91–99.7%-of-variance level.
- **No flight data, no TRL claim.** Nothing here is a certified interoperability standard, a reproduction of any agency’s internal budget, or a statement at technology readiness level above analysis. The deliverable is a proven theorem, one validated real-data decomposition, and a modelled design law built on them.

None of these caveats touches the proven content—the gauge lemma, the removability structure of the split, and the tolerance inversion—or the validated content—the real inter-ephemeris decomposition. Each marks where a real multi-provider deployment must supply flight data before any absolute budget number is trusted.

8 Conclusion

A multi-provider lunar PNT architecture needs a number that no interoperability standard yet publishes: the consistency tolerance, the maximum cross-provider frame-and-dynamics disagreement a user can absorb within a positioning budget. The LunaNet Interoperability Specification defers it to a still-unreleased applicable document—AD-5, the Lunar Reference System and LunaNet Reference Time System Standard (LNIS-TBD-AD0005)—which does not yet specify one. We have supplied it in three parts. A consistency-tolerance theorem shows the interop tax is a free-network gauge phenomenon—a datum common to all providers is invisible to a single provider and cancels in provider-versus-provider fusion, while the *differential* datum that two providers do not share is the tax—that splits, as a measured affine decomposition of the recovered rotation, into a reducible part a common frame removes and an irreducible part only a common ephemeris removes, and that inverts into a closed-form tolerance $\tau(B) = B/R_{\text{Moon}}$, binding on orientation, for a user budget B . The only real “multiple providers” that exist today—the three independent authoritative lunar ephemerides DE440, INPOP21a, and EPM2021—anchor the budget: their measured 2024–2025

geocentric-Moon disagreement is an orientation effect (a constant rotation explains 71–94% by amplitude, 91–99.7% by variance), splitting into a reducible frame-tie and an irreducible Moon-excess ($\approx 5\text{--}6$ nrad, a ≈ 2 m geocentric-Moon-state envelope at the Earth–Moon lever arm) for the JPL-versus-others pairs; this decomposition is validated against an independent SciPy reference to a tolerance below 10^{-3} . Folding the two together gives the design law: with an irreducible fraction of 0.687 (the irreducible rotation about twice the reducible), a common format-and-frame tag leaves the dominant ≈ 1.8 m geocentric-Moon-state floor—only millimetres for a surface user—and closing it requires mandating a common (or consistency-bounded) dynamical reference.

The design law is compact enough to state in one line: *for cross-provider lunar PNT, standardize the ephemeris, not just the format*. It gives interoperability programmes—LNIS AD-5, ESA Moonlight/LCNS, the CCSDS/IOAG cislunar standards—a concrete, real-data-backed requirement structure and a tolerance map, on an open, fully reproducible substrate. The theorem and the gauge lemma are proven; the inter-ephemeris decomposition is validated on real data; the system-level budget and tolerance map are explicitly modelled, because the providers are ephemerides and no two lunar-PNT providers fly. This paper is the second of a programme on lunar-PNT requirements science built on the same substrate as the companion datum-identifiability paper [3]; before any absolute budget number is trusted, a real multi-provider deployment must substitute flight data for the representative ephemeris analog used here. The convention-free content—an irreducible, orientation-type, Moon-specific disagreement between the real lunar dynamical references, removable by no coordinate convention—will carry over unchanged.

Data and code availability

The cross-provider interoperability substrate is part of the open-source `kshana` engine [4] (<https://github.com/ashfordeOU/kshana>; AGPL-3.0-only). The Helmert-difference fit, the rotation-only provenance decomposition, the reducible/irreducible split, the multi-provider budget, and the consistency-tolerance inversion are in the module `lunar_interop_budget` (functions `provenance_split`, `interop_budget`, and `consistency_tolerance`). Every number in this paper is regenerated by

```
cargo run --example p2_cross_provider_interop
```

which prints Part A (the per-pair rotation-only decomposition of Table 1), Part B (the multi-provider budget of Table 2), and Part C (the consistency-tolerance map of Table 3), directly from the committed engine and the vendored fixtures. The independent SciPy/NumPy oracle is `scripts/gen_interop_ref.py`, which reads the DE440/INPOP21a/EPM2021 kernels through IMCCE `calceph` and emits the reference decomposition; its output, the sampled geocentric-Moon and planetary-SSB positions, and the provenance notice are vendored at `tests/fixtures/inter_ephemeris/` (`moon_geo.csv`, `planet_ssb.csv`, `reference.json`, `NOTICE.md`). The ephemeris kernels themselves are not vendored (they are large and their redistribution terms vary); the notice records their public download URLs. The VALIDATED row—the engine’s decomposition matching the SciPy oracle to the stated tolerance—is asserted in the continuous-integration suite by `tests/lunar_interop_budget_reference.rs`, which fails the build on any mismatch. All frame, timescale, and sampling conventions are recorded in Appendix A and the vendored notice.

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The numerical substrate underlying this work is part of the open-source `kshana` engine. The lunar ephemerides differenced here are the property of their respective institutions (JPL, IMCCE, IAA RAS), gratefully acknowledged. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

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A Methods and modelled assumptions

This appendix records the ephemeris handling, frame and timescale conventions, the sampling and numerical method, and the frame-tie discrimination behind the results. All are consistent with the vendored provenance notice (Section 8).

A.1 Ephemeris provenance and access

Three independent authoritative ephemeris families are differenced, read through the IMCCE `calceph` library so that all three are accessed by a single interface in one frame and timescale:

- **DE440** (JPL/NASA), kernel `de440s.bsp` [12], from https://naif.jpl.nasa.gov/pub/naif/generic_kernels/spk/planets/de440s.bsp;
- **INPOP21a** (IMCCE / Observatoire de Paris), kernel `inpop21a_TDB_m100_p100_littleendian.dat` [5], from <https://ftp.imcce.fr/pub/ephem/planets/inpop21a/>;
- **EPM2021** (IAA RAS), kernel `epm2021.bsp` [13], from <https://ftp.iaaras.ru/pub/epm/EPM2021/SPICE/>.

Following the vendoring convention of the companion paper’s LLR fixtures [3], the kernels are *not* redistributed (they are large and their redistribution terms vary); only the sampled derived positions and the computed reference decomposition are vendored. Published scientific ephemeris positions are factual constants and are not copyrightable; the derived sample points are cited with full attribution to JPL, IMCCE, and IAA RAS.

A.2 Frame, timescale, and sampling

All positions are computed in the ICRF/J2000 frame and Barycentric Dynamical Time (TDB), so the three families are directly comparable without a frame or timescale conversion between them. Two position sets are sampled over the window 2024-01-01 to 2025-12-31 (TDB) at two-day cadence (366 epochs):

- `moon_geo.csv`: the geocentric Moon position (NAIF body 301 with respect to 399), in metres. This is the object whose disagreement is decomposed in Table 1; its metre-per-nanoradian lever arm is the Earth–Moon distance $r = 3.840 \times 10^8$ m.
- `planet_ssb.csv`: the Mercury, Venus, Mars, and Earth–Moon–barycentre positions with respect to the Solar System barycentre (NAIF bodies 1, 2, 4, 3 with respect to 0), in metres. These witness the common ICRF frame-tie rotation (Section A.4).

Both files, and the SciPy oracle, round all positions to six decimals (micrometre resolution) *before* the fit, so that the engine and the oracle consume byte-identical inputs. The validation therefore tests agreement of the two fitting *algorithms*, not the coincidence of two independent samplings; any sampling-level difference is removed by the shared rounded fixture.

A.3 Numerical method

The constant Helmert datum δ_{ab} of (1) is the least-squares solution of the stacked point-difference system with design matrix rows $\mathbf{J}(\mathbf{p}_i)$. We solve the centred normal equations—subtracting the per-provider centroid before assembling the system—with Jacobi (diagonal) preconditioning. Two reasons motivate this choice. First, centring removes the translation–rotation cross-coupling that otherwise inflates the condition number: for points at radius $\sim r$, the rotation and scale columns of $\mathbf{J}$ carry magnitudes $\sim r$ while the translation columns are $O(1)$, so the raw normal matrix has condition number $\sim r^2 \sim 10^{17}$; centring and Jacobi preconditioning (dividing each column by its norm) bring the columns to comparable scale and the condition number to $O(1)$, restoring double-precision accuracy. Second, for the rotation-only sub-fit (4) we use the closed-form singular value decomposition of the centred cross-covariance [2], which is the numerically stable route to the rotation and is the form the independent oracle also uses (so the two agree to the tight tolerance of Section 6.1). The general-purpose linear algebra follows standard practice [7].

A.4 Frame-tie discrimination and the Sun exclusion

The reducible/irreducible split requires separating a rotation common to all bodies (a frame-tie) from a rotation specific to the Moon (dynamics). We estimate the body-common rotation $\boldsymbol{\theta}_{\text{tie}}$ by fitting the same rotation-only model *separately* to each of the four witness bodies (Mercury, Venus, Mars, and the Earth–Moon barycentre), obtaining four per-body rotation vectors, and taking $\boldsymbol{\theta}_{\text{tie}}$ as their *component-wise median* (the median of the four x -components, and likewise for y and z). This is a deliberately robust but ad-hoc aggregate: the median resists a single discrepant witness, but it is *not* a joint fit across the witness bodies and *not* a weighted fit, and a different aggregation would shift the recovered $\boldsymbol{\theta}_{\text{tie}}$ —and hence the split—quantitatively (Section 7.3). The Moon-specific excess is then $\boldsymbol{\theta}_{\text{exc}} = \boldsymbol{\theta}_{\text{moon}} - \boldsymbol{\theta}_{\text{tie}}$, and the metre envelopes of Table 1 follow by mapping $\boldsymbol{\theta}_{\text{tie}}$ and $\boldsymbol{\theta}_{\text{exc}}$ through the Earth–Moon lever arm; because the two are in general non-parallel, these are worst-case upper bounds that do not sum to the raw disagreement.

The witness set is the four inner bodies Mercury, Venus, Mars, and the Earth–Moon barycentre. The Sun is deliberately *excluded*. The Sun’s position with respect to the Solar System barycentre

is dominated by the reflex motion imposed by the giant planets and is, by definition, tied to the barycentre convention itself; a difference in how two ephemeris families realize the barycentre therefore contaminates the Sun’s SSB position with a signal that is not a clean rigid frame rotation. Including it would bias the frame-tie estimate toward a barycentre-definition artifact rather than the ICRF orientation tie we intend to capture. The inner planets and the Earth–Moon barycentre, by contrast, have SSB positions dominated by their own orbits and provide cleaner witnesses of a common frame rotation. This choice is a MODELLED decision; Section 7.3 notes that a different witness set or weighting would shift the split quantitatively while leaving the qualitative Moon-excess dominance (for the JPL-versus-others pairs) intact, and that the convention-free content is the Moon-excess itself (Remark 1).